Dr. A. P. J. Abdul Kalam Technical University, Uttar Pradesh, Lucknow

UNDERGRADUATE COURSE

BACHELOR OF DESIGN

BATCH 2017-2018

PREFACE

Design is a profession of thinkers and visionaries, people who shape human experience of the future by learning from the past and the present. They're trained by education and practice to spot patterns, trends and gaps in people's day to day lives and gain insights from them. For these insights to be objective, meaningful and most importantly, actionable enough to evolve into ideas that improve human condition as well as the environment, a multidisciplinary field like design offers itself like a framework of effective problem solving.

This framework adopts practices from many other fields. The scientific method dictates the terms of collection and handling of information in the design practice. Technology drives the functional and engineering prudence of a design solution. Arts and humanities cover the characteristically emotional aspects of it. Hard economics dictates the terms of execution while culture and tradition influence the aesthetic and acceptability parts of a design solution.

With advances in engineering, designers strive to bring every technological breakthrough into the realm of human experience. They try to find new ways to use novel materials and processes to make incremental improvements in the quality of life in general. And now, they also continue to find effective ways to use the existing materials and processes in pursuit of averting further deterioration of the climate.

From the point of view of anthropology, as human behavior evolves and the society changes with it, designers try to gauge this evolution and offer tangible and non-tangible ways help people adapt to the changes induced by it.

Considering these and other aspects of the design profession, the process to train young designers becomes inherently multidisciplinary. It promotes lateral thinking and drives innovation on one hand, while providing tools to further accelerate the structured decision making process to streamline conventional problem solving on the other.

Design students are primed for being a part of the highly skilled talent pool for organizations. They are trained to become entrepreneurs, and they are taught to become scholars for various domains. Hence, an honorable Bachelor's of Design program must develop a set of skills and instill some virtues in its graduates to be able to contribute to the culture and economy at a level proportionate to their talent and training.

First, an **industry standard skill set** is an absolute requirement for employability in private or public sector in leading design positions. This skill set includes a certain degree of emotional intelligence to connect and empathize with the end users, analytic skills to distill actionable insights from the market and industry for a timely initiation of a design project, visualization skills to see and present the problems and solutions as they should be, and communication skill to effectively articulate the design process to the stakeholders.

To develop this skill set, the curriculum must include a sufficient workload of skill development courses that includes topics like Representation Techniques, technical drawing and drafting, 3D modeling, model making, materials and process, workshop and manufacturing technologies, technical Studies (Applied Sciences, Mechanics, Electronics, Programming), prototyping, elements of visual design, elements of form, color and composition, ergonomics, research methodologies, design management and system thinking in design. Given the importance of these skills in the profession, the courses covering them constitute the **Core** of the curriculum.

Second, the **qualifications** for enrollment in recognized Master's programs worldwide in Design, Applied Sciences and Humanities are decided by the academic credentials earned by a student during his degree program. The design courses typically exceed the undergraduate and postgraduate norms from the perspective of most of the nationally as

well as globally recognized evaluation systems as the number of topics covered and required contact hours to do so is significantly higher than a typical Technical or Engineering Degree program. However, it is not just the academic transcript that matters in a design degree, it is the quality and potential impact of the design projects and degree projects of the students which culminate in their portfolios and showcase their personal capacity to learn as well as the institution's capacity to teach and avail resources to visualize and materialize said projects. Students who pursue higher education are aiming to be specialists of a respective sub-domains of design, so their interests in a particular field of design are reflected in the courses they choose to build their theoretical foundation. Subjects offering the starting point for higher education programs form the **compulsory and elective foundation** part of the curriculum.

Third, the **analytic** skills are required for design students to be an active part of the Research and Development pool of academia as well as the economy. Courses like Design management and Research methodology offer such skills, hence should be included in the core of the curriculum.

Fourth is the **entrepreneurship** and management skills that the graduates require for starting their own innovation driven ventures. Since design as a profession transcends the boundaries of multiple domains of expertise, the true potential of some designers is realized when they capitalize upon their skills by offering them to a multitude of domains as entrepreneurs. Hence, they should have a basic level of awareness about starting up a business and its sustainable operation. This awareness also helps them in understanding the needs and limitations of the organizations they work for.

Fifth is the **cultural sensibility** that helps the designers judge the potential acceptability of their work in society. It is also essential so that solutions adopted from other cultures can be translated into more acceptable local versions.

Sixth, **socio-economic sensibility** is essential for building social ventures concerned with public affairs. It also helps the design practitioners in being aware of concepts like corporate social responsibility, grass root issues etc. Without being sensitive to customs, traditions and the local ecosystems, designers cannot produce truly inclusive solutions.

Courses like design and ethics, field study parts of the curriculum projects, workshops and short term modules prepared by experts in related fields, and periodic documentation filing assignments help the students develop this aspect of their professional persona.

Seventh, **sensibility for environmental concerns** encourage designers to contribute for the stabilization of rapidly declining climate quality of the planet. Introduction of basic environmental sciences sensitizes designers towards the potential environmental impact their work may have. Given the impact on future generations, and the repercussions faced current generation due to disregard of natural cycles, this subject becomes crucial and mandatory part of the curriculum.

And eighth, a **global perspective** that helps the graduates in promoting India's design heritage in other parts of the world and bringing in the best practices of other economies and cultures into India's ecosystem. Academic exchanges across the universities and promotion of institutional publications like studies and other body of work of the students and the faculty helps in this regards.

With these considerations in mind, a model curriculum of Bachelor's degree in Design is offered in this document with an intention to prepare graduates who can successfully render their services to the society, industry and the environment for everyone's benefit and their personal growth in a profession that is challenging, emotionally and physically taxing yet exciting and fulfilling, philosophical yet pragmatic and emotional yet technology driven.

Curriculum (2017-18 Batch)

SEMESTER I						
S. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ADS 101	INTRODUCTION TO DESIGN	1	0	6	4
2	ADS 102	REPRESENTATION TECHNIQUES	0	0	4	2
3	ADS 103	ELEMENTS OF DESIGN	1	0	8	5
4	ADS 104	APPLIED SCIENCES FOR DESIGN	2	0	6	5
5	ADS 105	TINKERING STUDIO	1	0	6	4
6	AME 111	ENGINEERING DRAWING	1	0	2	2
7	AEE 101	ELECTRICAL SCIENCES	1	0	2	2
		TOTAL CREDITS FOR SEMESTER I				**24**
SEMESTER II						
S. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ADS 111	FORM ORDER AND STRUCTURE	1	0	8	5
2	ADS 112	PRINCIPLES OF VISUAL DESIGN	1	0	8	5
3	ADS 113	VISUALIZATION AND ILLUSTRATIONS	0	0	10	5
4	ACS 101	INTRODUCTION TO COMPUTING	3	0	0	3
5	ACS 110	COMPUTING LABORATORY	0	0	4	2
6	AEE 102	BASIC ELECTRONICS LABORATORY	0	0	4	2
7	AME 101	ENGINEERING MECHANICS	1	0	2	2
		TOTAL CREDITS FOR SEMESTER II				**24**

SEMESTER III						
S. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ADS 211	ARCHITECTURAL STUDIES IN DESIGN- I, SPACE AND HABITAT	2	0	4	4
2	ADS 212	COMMUNICATION STUDIES IN DESIGN- II	3	0	0	3
3	ADS 213	MODEL MAKING AND HAND TOOLS WORKSHOP	1	0	6	4
4	ADS 214	SURFACE MODELLING IN COMPUTER AIDED DESIGN	1	0	8	3
5	RVE 301	UNIVERSAL HUMAN VALUES AND PROFESSIONAL ETHICS	3	0	0	3
6	ADS 271	DESIGN DOCUMENTATION- I	0	0	0	2
7	ADS 281	DESIGN PROJECT- I, SIMPLE PRODUCT DESIGN	0	0	10	5
8	---	CYBER SECURITY	2	0	0	0
		TOTAL CREDITS FOR SEMESTER III				**24**
SEMESTER IV						
S. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ADS 221	DESIGN MANAGEMENT- I, CONSUMER PSYCHOLOGY	2	0	2	3
2	ADS 222	DESIGN THINKING	1	0	6	4
3	ADS 223	SOLID MODELLING IN COMPUTER AIDED DESIGN	1	0	6	4
4	---	ENGLISH LANGUAGE AND TECHNICAL WRITING	2	0	2	3
5	ADS 261	NATURE AND FORM	0	0	10	5
6	ADS 291	DESIGN PROJECT- II, DISPLAY AND CONTROL DESIGN	0	0	10	5
7	---	ENVIRONMENTAL SCIENCE	2	0	0	0
		TOTAL CREDITS FOR SEMESTER IV				**24**

SEMESTER V						
S. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ADS 311	CREATIVE NARRATION	2	0	0	2
2	ADS 312	HUMANITIES & SOCIAL STUDIES	2	0	0	2
3	ADS 313	DESIGN MANAGEMENT- II, BUSINESS COMMUNICATION	2	0	0	2
4	ADS 314	DESIGN RESEARCH METHODOLOGY- I, QUANTATIVE RESEARCH	2	0	0	2
5	ADS 315	ARCHITECTURAL STUDIES –II, CONNECTIVITY AND MOBILITY	1	0	4	3
6	ADS 331	ELECTIVE- I	1	0	2	2
7	ADS 332	ELECTIVE- II	1	0	2	2
8	ADS 351	DESIGN WORKSHOP	0	0	8	4
9	ADS 381	DESIGN PROJECT- III, SOFTWARE USER INTERFACE DESIGN	0	0	10	5
10	---	CONSTITUTION OF INDIA	2	0	0	0
		TOTAL CREDITS FOR SEMESTER V				**24**
SEMESTER VI						
S. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ADS 321	PRODUCT BRANDING AND IDENTITY	2	0	4	4
2	ADS 322	DESIGN RESEARCH METHODOLOGY- II, APPLIED STATISTICS AND INFOGRAPHICS	3	0	2	4
3	ADS 323	MATERIAL AND PROCESSES IN DESIGN- I, UNCONVENTIONAL MANUFACTURING	1	0	8	5
4	ADS 341	ELECTIVE- III	1	0	2	2
5	ADS 342	ELECTIVE- IV	1	0	2	2
6	ADS 361	CONTROL SYSTEM IN DESIGN	0	0	4	2
7	ADS 391	DESIGN PROJECT- IV, TECHNICALLY COMPLEX PRODUCT DESIGN	0	0	10	5
8	---	ESSENCE OF INDIAN TRADITION KNOWLEDGE	2	0	0	0
		TOTAL CREDITS FOR SEMESTER VI				**24**

SEMESTER VII						
S. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ADS 411	PROFESSIONAL PRACTICE IN DESIGN	2	0	0	2
2	ADS 412	FORM IN FOUR DIMENSION	2	0	10	5
3	ADS 413	MATERIAL AND PROCESSES IN DESIGN –II, PRODUCTION PLANNING	2	0	2	3
4	ADS 431	ELECTIVE- V	1	0	2	2
5	ADS 471	INDUSTRIAL TRAINING	0	0	0	3
6	ADS 472	DESIGN DOCUMENTATION-II	0	0	0	4
7	ADS 481	DESIGN PROJECT- V, SYSTEM THINKING IN DESIGN	0	0	20	5
		TOTAL CREDITS FOR SEMESTER VII				**24**
SEMESTER VIII						
S. No.	**Course Code**	**Course Name**	**L**	**T**	**P**	**C**
1	ADS 499	DESIGN DEGREE PROJECT	0	0	48	**24**
		TOTAL CREDITS FOR SEMESTER VIII				**24**
		TOTAL CREDITS				**192**

DESIGN THINKING – I

ADS101–INTRODUCTION TO DESIGN

Course Outcome

Discovery of design as a profession

Awareness about psychology of visual perception

Awareness about historical perspective of aesthetics

Understanding of materials an processes

Ability to define problems

Unit 1: Introduction to Design

Definition and understanding of the design process. A brief history of Design, evolution of industrial design as a profession.

Unit 2: Principles of Gestalt

Basic understanding of visual perception. Discussion and demonstration of the way humans make a greater sense out of combination of simple or complex curves and shapes.

Unit 3: Introduction to Product Design

Definitions of material, process and product. History and examples of product design. Fundamentals of Product development cycle. Understanding the concept of end user. Defining user groups. Understanding the concept of Market. Defining the market gap.

Unit 4: Design Concepts and concerns

Problem identification and definition. Scenario Building. Mind Maps and Metaphors. Definition of a solution. Ideation, Conceptualization and representation of solutions.

Unit 5: Introduction to system thinking

Definition of a system. Understanding the concepts of component, connection, transaction and transformation. Understanding the boundary and boundary conditions. Understanding the concepts of input, activity, response and output. Introduction to Static and dynamic maps.

References

T. Hauffe, Design: A Concise History, Laurence King Publishing, 1998
P.B. Meggs, A History of Graphic Design, Library of Congress Cataloging, 1998
Alan Pipes, Foundation of Art and Design, Lawrence King Publishing, 2008
Bryan Peterson, Design Basics for Creative Results, How Design Books, 2003
Don Norman, Emotional Design, Why We love (or Hate) Everyday Things, Basic Books, 2003
Don Norman, Design of Everyday Things, Basic Books, 2014
E. Pilloton, Design Revolution: 100 Products That Empower People, Metropolis Books, 2009
Doblin jay, 100 Great Product Designs
E. Lupton and J.A. Miller, The ABC's of Bauhaus: The Bauhaus and Design Theory, Princeton Architectural Press, 1991
Manmohan Ghosh, The Natyashastra: A treatise on Hindu Dramaturgy and Histrionics as ascribed to Bharata Muni, Vol. I, The Royal Asiatic Society of Bengal, Calcutta, 1959

ADS102– REPRESENTATION TECHNIQUE

Course Outcome

Ability to create compelling and detailed line drawings of real or imaginary objects in short time.

Ability to effectively explain an idea through visual language

Ability of create simplified representations.

Ability to generate novel patterns and textures

Basic understanding of composition in graphic design

Unit 1: Introduction to Drawing

Introduction to the skills of drawing and sketching, methods and techniques of penciling;

Unit 2: Drawing from still objects

Drawing a still life using different objects of different sizes and shapes,

Exploring Material Properties.

Unit 3: Perspective and Composition

Use of drawing as a means of developing creative thinking, develop and understanding of perspective and composition.

Unit 4: Sketching 1

Different approaches to explore and record daily observations, ideas, imagination and concepts.

Drawing from memory and imagination.

Unit 5: Sketching 2

Representing nature; Figure drawing.

References

William J. Mitchell, The Nascent Medium, The MIT Press, 1992
Dalley Terence, The complete Guide to Illustrations and Design, Phaidon, 1980
Maurello Ralphs, The complete air brush, Leon Amiel Publishers Inc. NY, 1980
Schmid & Schmid, Handbook of Graphic Presentation, Roland Press, 1979
Herdeej, Graphic Diagrams, Graphic Press, 1976
Langford Michael, Advance Photograph, Focal Press, 1974
Hedgeco John, The air of colour photography, Michel Beazley, 1981

ADS103– ELEMENTS OF DESIGN

Course Outcome

To learn basic elements of visual design and its grammar. Pattern recognition, abstraction and construction.

Unit 1: Point/Dot, Lines and curves

Characteristics of point/dot. Connotations, examples of use in arts, visual communication and industrial design. Characteristics of a line. Line Quality. Curves, qualities and characteristics of curves. Visual and emotions interpretations.

Unit 2: Shapes and Patterns

Characteristics of a shape, concepts of positive and negative space, types of shapes. Developing patterns by repetition of points, dots, rectilinear elements, curvilinear elements, Shapes. Regular and Irregular Patterns. Fractals.

Unit 3: Texture

Understanding of flat surface. Material and process based textures, construction based textures, pigmented textures.

Unit 4: Area, Volume and Form (Primary)

Gradation on Flat Surface, Introduction of highlights and shadows. Primer for elements of form course

Unit 5: Colour (Primary)

Introduction to colors, pigment and light, additive and subtractive models. Shades of Greys. Understanding warm and Cold Greys.

References

Samara Timothy, Design Elements, 2nd Edition: Understanding the rules and knowing when to break them, Rockport Publishers, 2014
Evans Poppy, Thomas Mark A., Exploring the Elements of Design, Delmar Cengage Learning, 2012
Beech R., Origami – The Complete Guide to the Art of Paper Folding, Lorenz Books, 2001
Wong W., Principles of Two Dimensional Design, John Wiley & Sons, 1972
White Alex W., The Elements of Graphic Design, Allworth Press, 2011
Itten J., The art of colour: the subjective experience and objective rationale of colour, John Wiley and Sons. 1974
Sherin, A., Design Elements, Color Fundamentals: A Graphic Style Manual for Understanding How Color Impacts Design, Beverly, Mass: Rockport Publishers, 2011

ADS104 – APPLIED SCIENCE FOR DESIGN

Unit 1: Design in natural world

Evolution of species' & evolution and metamorphosis as design refinement, mechanisms and bio-mechanisms & bio-circulations.

Unit 2: Basic robotics

Genesis of robotics and differentiation, nomenclature of robotic parts, applications of robots, sensors used in various robots. Studio hours involve development of a robotic mechanism in a chosen life form.

Unit 3: Basic Aerodynamics

Bernoulli's principle, Fluid dynamics conservation laws, static equilibrium and trim.

Unit 4: Basic Pneumatics & Hydraulics

Comparisons of electrical, hydraulic and pneumatic systems, Pascal's law, hydraulic pumps & air compressors, hydraulic and pneumatic accessories

Unit 5: New Materials

Smart materials, Nano materials, their application and examples of usage.

Course Outcome

Awareness about natural ecosystem.

Ability to assess the working mechanism of basic objects.

References

Moshe Shoham; A Textbook of Robotics 1 Basic Concepts, Eshed Robotec 1984
Myke Predko; 123 Robotics Experiments for Evil Genius, 2004
Garratt J.; Design and Technology, Cambridge University Press, 1996
Henry T Brown; 507 Mechanical movements, Brown Coombs & Company
Michael F Ashby, Daniel L Schodek, Paulo J Ferreira; Nano materials, Nano technology & design, Elsevier Ltd, 2009
Duane Knudson; Fundamentals of Biomechanics, Springer Science+Business Media, LLC, 2007
Jared Diamond; Guns Germs and Steel, W.W. Norton 1997

ADS105– TINKERING STUDIO

Course Outcome

Ability to identify working mechanisms that make a product.

Building a creative self.

Unit 1: Linkages and Joints

Introduction of links, mechanisms, inversions, joints etc. (natural and artificial), its applications in various domains.

Unit 2: Cams and Gears

Understanding of the concept of Locus, Introduction of cams and gears, Gear ratios, mechanical advantage, types of gear and cams, and its applications.

Unit 3: Lateral thinking

Lateral thinking by creating puzzle / game board's designs and working solutions using newer or available technology / DIY kits. Creative thinking by solutions to an existing real world problem.

References

Sclater, N., &Chironis, N. P. (2001). Mechanisms and mechanical devices sourcebook (Vol. 3). New York: McGraw-Hill.
Ratan, S. S. Theory of machines. S Chand.
Edward de Bono, Lateral Thinking, Penguin UK, 2010
Garratt J., Design and Technology, Cambridge University Press,1996
Don Norman, The Design of Everyday Things, Basic Books, 2014
Martin Gardner; 100 Perceptual puzzles, 1995 Barnes & Noble Books

AME111– ENGINEERING DRAWING

UNIT1: ENGINEERING DRAWING

Lines, Lettering, Sketching, Principle of Dimensioning, Basic geometrical constructions, Scales, Engineering Curves.

Course Outcome

Ability to create detailed engineering drawings for production

UNIT 2: ORTHOGRAPHIC PROJECTIONS

Pictorial view, Terminology, First angle projection and its features, Third angle projections and its features, Symbols, Section lines or hatching, Conversion of pictorial view into orthographic view.

Unit 3: Projection line

Introduction, Orientation of a line, Projections of a line located at different locations, Projections of a line in different angles.

Unit 4: Sectional Views

Introduction, Terminology, Types of section planes, Section by a plane perpendicular to VP, HP and both.

Unit 5: Computer Aided Drafting

Introduction to Software and its tools, computer generated technical drawing and assembly.

References

A.J. Dhananjay, Engineering Drawing, TMH, 2008
N D Bhatt and V M Panchal, Engineering Drawing, 43rd Ed., Charator Publishing House,2001
M B Shah and B C Rana, Engineering Drawing, 2nd Ed., Pearson Education, 2009
T E French, C J Vierck and R J Foster, Graphic Science and Design, 4th Ed., McGraw Hill, 1984.
W J Luzadder and J M Duff, Fundamentals of Engineering Drawing, 11th Ed., PHI, 1995
K Venugopal, Engineering Drawing and Graphics, 3rd Ed., New Age International, 1998

AEE101 –ELECTRICAL SCIENCES

Course Outcome

Ability to understand basic electrical & electronic concepts & their functioning.

Unit 1:

Circuit Analysis Techniques: Circuit elements, Simple RL and RC Circuits, Ohm's law, Kirchhoff's laws, Nodal Analysis, Mesh Analysis, Linearity and Superposition, Source Transformations, Thevnin's and Norton's Theorems, Sinusoidal Forcing Function, Complex Forcing Function, Phasor Relationship for R, L and C, Impedance and Admittance, Phasor Diagrams, Response as a function of ω.

Unit 2:

Diodes and Transistors: Semiconductor Diode, Zener Diodes, Rectifier Circuits, Wave Shaping Circuits, Bipolar Junction Transistors, Field-Effect Transistors, Transistor Biasing, Transistor Small Signal Analysis, Transistor Amplifiers.

Unit 3:

Operational Amplifiers: Op-amp Equivalent Circuit, Practical Op-amp Circuits, DC Offset, Constant Gain Multiplier, Voltage Summing, Voltage Buffer, Controlled Sources, Instrumentation Circuits, Active Filters and Oscillators.

Unit 4:

Logic Gates and Combinational Circuits: Number Systems and Codes, Logic Gates, Boolean Theorems, DeMogan's Theorems, Sum-of Product Form, Algebraic Simplification, Karnaugh Map Method, Parity Generator and Checker, Inhibit Circuits. Sequential Circuits and Arithmetic Circuits: NAND and NOR gate Latches, S-C Flip-Flop, JK Flip-Flop, D Flip-Flop, Data Storage, Serial Data Transfer, Frequency Division and Counting, Binary Addition, 2's Complement System, Full Adder, BCD Adder.

Unit 5:

Transformers and AC Machines: Ideal Transformer, Circuit Model of Transformer, Determination of Parameters of Circuit Model of Transformer, Voltage Regulation, Efficiency, Three Phase Induction Motor, Three Phase Synchronous Generator, Induced Voltage, Electromagnetic Torque, Equivalent Circuit of Three Phase Induction Motor, Torque Speed Characteristic.

Fractional-kW Motors and DC Machines: Single Phase Induction Motors, Characteristics and Typical Applications, Stepper Motors, Construction Features, Methods of Operations, DC Generator and DC Motor Analysis, Methods of Excitation, Speed Torque Characteristics and Speed control of DC Machines.

References

W.H. Hayt and J.E. Kemmerly, Engineering Circuit Analysis; McGraw-Hill, 1993
R.J. Smith and R.C. Dorf, Circuits, Devices and Systems; John Wiley & Sons, 1992
R.L. Boylestad and L. Nasheisky, Electronic Devices and Circuit Theory; PHI, 6e, 2001
R.J. Tocci, Digital Systems; PHI, 6e, 2001
V. Del Toro, Electrical Engineering Fundamentals; PHI, 1994.
D.P. Kothari and I.J. Nagrath, Basic Electronics; McGraw-Hill, 1993
D.P. Kothari and I.J. Nagrath, Basic Electrical Engineering; McGraw-Hill, 1991

ADS 111 –FORM, ORDER AND STRUCTURE

Course Outcome

Understanding and appreciation of 3D & natural forms.

Ability to construct & manipulate forms.

Unit 1:

Introduction to the geometry of platonic solids and study of their interrelationship;

Unit 2:

Derivation of Archimedean solids through truncation of regular solids;

Unit 3:

Boolean solids; Order and structure- interrelationship of 2D and 3D forms;

Unit 4:

Construction of solids using paper; Form manipulation; Form transition

Unit 5:

Expressive form, combinatory forms and topology of 3D forms

References

W. Wong; Principles of Two Dimensional Design, John Wiley & Sons, 1972
K. Critchlow; Order in Space: A design Source Book, Thames and Hudson, 1969
E. H. Gombrich; The Sense of Order, Phaidon Press, 1994
R. Beech, Origami: the complete guide to the art of paper folding, Lorenz Books, 2001
C. Akner-Koler, Three Dimensional Visual Analysis, Konftfack, Sweden, 1994

ADS 112 –PRINCIPLES OF VISUAL DESIGN

Course Outcome

Ability to understand & develop visual design language.

Understanding the emotional impact of colours on people & create compelling design.

Unit 1:

Study of visual principles – Balance: Formal and informal balance;

Unit 2:

Harmony / Unity; Emphasis / Focus

Unit 3:

Rhythm: different types of rhythm; Pattern; Contrast; Scale and proportion;

Unit 4:

Visual hierarchy; Visual Order and Chaos; Positive – Negative space;

Unit 5:

Tessellation: Regular and Semi-regular tessellation, modular tessellations

References

Gonnella, Rose., Navetta, Christopher and Friedman, Max. Design Fundamentals: Notes on Visual Elements and Principles of Composition. Peachpit Press, 2015

Lidwell, William and Holden, Kritina. Universal Principles of Design. Rockport Publishers, 2010

Pipes, Alan. Foundations of Art and Design. Laurence King Publishing, 2008

Poopy Evans, Mark Thomas. Exploring the Elements of Design. Thomson Delmar Learning, 2008

Bryan, Peterson. Design Basics for Creative Results, How Design Books, 2003

Wong, Wucius. Principles of Two Dimensional Design. Canada: John Wiley & Sons, 1972

ADS113 – VISUALIZATION AND ILLUSTRATIONS

Course Outcome

Ability to quickly visualize ideas and execute visual design explorations.

Ability to create compelling visuals of ideas before they come to reality for every stakeholder's proper understanding.

Unit 1:

Introduction to different medium, tools and instruments to create illustrations;

Unit 2:

Understanding basic principles of perception including depth and its representation in terms of reflections, rotations, sciography
Inter-relationship of visual forms in terms of size, scale and overall proportion.

Unit 3:

Conceptualizing ideas into an illustrative format like sections, cut-away and exploded views
Exploring the area of realistic to experimental illustrations.

Unit 4:

Nature drawing - to study form, structure and various shapes;
Representation of 3-dimensional forms;

Unit 5:

Introduction to basics of drawing techniques in the digital format;
Transforming works form sketching to digital format, using Google sketchup

References

T. C. Wang, Pencil Sketching, John Wiley & Sons,1997,
D.K, Francis, Design Drawing, John Wiley & Sons,1998

ACS101 -INTRODUCTION TO COMPUTING

Course Outcome

Understanding basics of modern computing and its implementation in Design.

Unit 1:

The von Neumann architecture, machine language, assembly language, high level programming languages, compiler, interpreter, loader, linker, text editors, operating systems, flowchart;

Unit 2:

Basic features of programming (Using C): data types, variables, operators, expressions, statements, control structures, functions; Advanced programming features: arrays and pointers, recursion, records (structures), memory management, files, input/output, standard library functions, programming tools,

Unit 3:

Fundamental operations on data: insert, delete, search, traverse and modify;
Fundamental data structures: arrays, stacks, queues, linked lists; Searching and sorting: linear search, binary search, insertion-sort, bubble-sort, selection-sort, radix-sort, counting-sort; Introduction to object-oriented programming;

References

Programming in ANSI C by E.Balagurusamy, Seventh Edition, McGrawHill,
A Book on C by Ira Pohl, Al Kelly

ACS110– COMPUTING LABORATORY

Programming Laboratory will be set in consonance with the material covered in introduction to computing. This will include assignments in a programming language like C.

References

Programming in ANSI C by E. Balagurusamy, Seventh Edition, McGrawHill,

Course Outcome

Readiness for programing control systems.

Readiness for building and testing software solutions.

AEE102 – BASIC ELECTRONICS LABORATORY

Course outcome

Readiness to build control system hardware for products.

Readiness to build powertrains in products.

Experiments using diodes and bipolar junction transistor (BJT): design and analysis of half -wave and full-wave rectifiers, clipping circuits and Zener regulators, BJT characteristics and BJT amplifiers; experiments using operational amplifiers (op-amps): summing amplifier, comparator, precision rectifier, a-stable and mono-stable multi-vibrators and oscillators; experiments using logic gates: combinational circuits such as staircase switch, majority detector, equality detector, multiplexer and de-multiplexer; experiments using flip-flops: sequential circuits such as non-overlapping pulse generator, ripple counter, synchronous counter, pulse counter and numerical display..

References

A. P. Malvino, Electronic Principles, Tata McGraw-Hill, New Delhi, 1993.
R. A. Gayakwad, Op-Amps and Linear Integrated Circuits, PHI, New Delhi, 2002.
R.J. Tocci, Digital Systems, 6th Ed., 2001

AME101– ENGINEERING MECHANICS

Unit 1: Basic principles of mechanics

Equivalent force system; Equations of equilibrium; free body diagram; Reaction; Static indeterminacy. Difference between trusses, frames and beams, Assumptions followed in the analysis of structures; 2D truss; Method of joints; Method of section; Frame; Simple beam; types of loading and supports; Shear Force and bending Moment diagram in beams; Relation among load, shear force and bending moment. Dry friction; Description and applications of friction in wedges, thrust bearing (disk friction), belt, screw, journal bearing (Axle friction); rolling resistance.

Course outcome

Theoretical understanding of how mechanisms and mechanical forces behave.

Ability to create simplified versions of complex mechanisms.

Ability to rapidly test mechanical solutions.

Unit 2: Virtual work and Energy method

Virtual Displacement; Principle of virtual work; Applications of virtual work principle to machines; Mechanical efficiency; Work of a force/couple (springs etc.); Potential energy and equilibrium; stability.

Unit 3: Centre of Gravity and Moment of Inertia

First and second moment of area; Radius of gyration; Parallel axis theorem; Product of inertia, Rotation of axes and principal moment of inertia; Moment of inertia of simple and composite bodies. Mass moment of inertia.

Unit 4: Kinematics and Kinetics

Kinematics of Particles; Rectilinear motion; Curvilinear motion; Use of Cartesian, polar and spherical coordinate system; Relative and constrained motion; Space curvilinear motion; Kinetics of Particles; Force, mass and acceleration; Work and energy; Impulse and momentum; Impact problems; System of particles; Kinematics and Kinetics of Rigid Bodies; Translation; Fixed axis rotational; General plane motion; Coriolis acceleration; Work energy; Power; Potential energy; Impulse-momentum and associated conservation principles; Euler equations of motion and its application.

Unit 5: Practical Mechanics

Exploring mechanical properties of model making materials through experiments; Introduction to DIY culture; Building mechanisms using workable materials; Exploring degrees of freedom, joints and links through experiments. 4 Bar Mechanisms, Grash of Criterion, Crank-Rocker, Double Crank, Double Rocker, and Slider-Crank mechanisms; Bio Mechanics; Introduction to the mechanical aspect of bio-mimicry, skeletal mechanisms of animal limbs, skeletal mechanisms of human limbs, structural characteristics of plants, structural characteristics of exoskeletons; Introduction to Control Systems; introduction to mechanical switches, power trains and experimentation with actuators and motors; Introductions to Arduino and Raspberry Pie;

References

I.H. Shames, Engineering Mechanics: Statics and Dynamics, 4th Ed., PHI, 2002

F. P. Beer and E. R. Johnston, Vector Mechanics for Engineers, Vol I - Statics, Vol II – Dynamics, 3rd Ed., Tata McGraw Hill, 2000

J. L. Meriam and L. G. Kraige, Engineering Mechanics, Vol I – Statics, Vol II – Dynamics, 5th Ed., John Wiley, 2002

R. C. Hibbler, Engineering Mechanics, Vols. I and II, Pearson Press, 2002

S. Timoshenko and D. H. Young, Engineering Mechanics, McGraw Hill, 1956

ADS211–ARCHITECTURAL STUDIESIN DESIGN–I, SPACE AND HABITAT

Course outcome

Understanding the concept of space and Place from a designer's point of view

Ability to differentiate between qualities of built and natural spaces

Understanding human behaviour in isolated and shared spaces

Understanding the need to utilize spaces for gratification

Unit 1: Space

Concept of Occupation; occupied spaces; unoccupied spaces; Built Spaces; natural Spaces; Landscapes; Arenas; Concept of boundary; Physical Boundaries; Imaginary boundaries; Interior Space;
Lighting and illumination, Climatology;
Concept of Play; Recreation; Comfort and Safety parameters; Activity Oriented Spaces;

Unit 3: Evolution of Spaces

Visual concepts of Area and Volume, history of built spaces, history of settlements;
Concept of Home; Shelters; Dwellings; Settlements; Solitary living; Co Habitation; Cooperative living; Adaption; Structure and Dynamics of Built Spaces; Residential; Commercial and Industrial Spaces;
Evolution of agrarian society; prehistoric settlements; Evolution of rituals and customs; Rural Sanitation; Rural connectivity; Migration patterns;

Unit 4: Visual Design in Spaces

History of Aesthetics in architecture; Hedonistic Vs Ascetic design of spaces; Design movements (-isms e.g., Modernism, post modernism, Memphis, art deco)

Unit 5: Utilitarian Spaces

Origin of commerce; commercial spaces; High density habitats; industrialization; post industrialization; Urban Housing;
Fundamentals of interior design and exhibitions; Public Spaces, Mobility oriented spaces;

References

Various, A Dictionary of Color Combination (Sanzo Wada's works), Seigensha Art Publishing, 2011
Charles Correa, A Place in the Shade, Penguin India, 2010
Setha M. Low, Denise Lawrence-Zúñiga, Anthropology of Space and Place: Locating Culture (Wiley Blackwell Readers in Anthropology) , Wiley-Blackwell , 2003
Nick Dunn, Architectural Model making 2nd Edition, Thames & Hudson, 2014
Katherine S. Willis, Net spaces: Space and Place in a Networked World, Routledge, 2017
Meg Boulton, Jane hawkes, Place and Space in the Medieval World, Routledge, 2017

ADS212–COMMUNICATION STUDIES IN DESIGN – II

Course Outcome

Understanding the mechanisms of human conversation

Understanding persuasive dialogue

Using persuasion in visual language

Unit 1: Information

Processing information – Coding & Decoding; Sender, Channel and Receiver; Signs and their meanings in Indian cultures

Unit 2: Communication

Study of relationships between Signifier, Signified and context, Denotation and Connotation; Communicating through gestures, voice, type and visuals; Designing visual messages to send meanings

Unit 3: Introduction to Semiotic Perspective

Goals of Semiotic Analysis; Sign: Concept and Types; Codes: Concepts, Types and Sharing; Process of Signification: Connotation and Denotation

Unit 4: Semiotic Interpretations and Culture

Metaphors; Myths: Concept and Debates; Communication as Text / Discourse; Ideology: Link to Meaning Making

Unit 5: Introduction to Rhetoric Perspective

Origin and Evolution; Functions of Rhetoric; Key Elements of Rhetoric; Introduction to Indian Thoughts on Rhetoric

Unit 6: Rhetoric Presentation and Effects

Rhetorical Schemes and Devices; Elements of Rhetoric Presentation; Analyzing Rhetorical Presentation; Making Rhetorical Presentation

References

Ronald H. Forgus, Perception, The basic process in cognitive development, USA, McGraw-Hill 1996
Arthaya, Seminar on Visual semantics, IDC, IIT Bombay 1992
Carey, J. W. (1989). Communication as culture: Essays on media and society. Boston: Unwin Hyman
Ghanekar, A (1998) Communication skill for effective management. Everest Publishing House. Gilligan, Pune
Fiske, J. (1982). Introduction to communication studies. London, Angleterre: Methuen
Schlenker, B. R. (1980). Impression management: The self-concept, social identity, and interpersonal relations. Monterey, Calif: Brooks/Cole Pub. Co.

ADS213 – MODEL MAKING AND HAND TOOLS WORKSHOP

Course Outcome

Understanding of the behaviour and properties of plastic components

Understanding of deformable materials

Finding out biodegradable replacements of polluting materials

Unit 1: Introducing Plastic Polymers

Properties and usage of thermoplastics and thermosetting plastics.

Unit 2: Plastics

Process of selection and applications of plastics for engineering and consumer products.

Design limitations and specific advantages of plastic molding processes. Property change on recycled plastics.

Environmental impact of disposable plastic products, recycling methods of different types of synthetic polymers, biodegradation, UV degradation

Unit 3: Elastomers

Properties, processing and use of natural and synthetic Rubber

Unit 4: Ceramics

Properties, Processing and use of ceramics and glass.

Unit 5: Natural materials

Properties of natural materials like wood, bamboo, cane, leather, cloth, jute and paper and their use at craft and industry. Workshop practice of woodworking and wood carving.

Preferences

J. Garratt, Design and Technology, Cambridge University Press, UK, 20004
R. Thompson, Manufacturing processes for design professionals, Thames & Hudson, London 2007
Michael Ashby and Kara Johnson, Materials and Design: The Art and Science of Material Selection in Product Design, Butterworth Heinemann, 2002

ADS214–SURFACE MODELLING IN COMPUTER AIDED DESIGN

Course Outcome

Ability to create Engineering CAD ready Surface models

Ability to conduct design iterations in CAD software

Unit 1: Points

Point Cloud data Handling; Fitting curves and surfaces through point Clouds; Interpolation basics

Unit 2: Curves

Scanning and tracing Sketches; Curve hierarchy, ISO-Curves; End Tangents; Multi Span Bezier Curves;

Unit 3: Surface Construction

Tolerances; open and closed surfaces; trimmed and untrimmed surfaces;

Unit 4: Freeform surface modeling

Sculpting with Control vertices; Sculpting with ISO-Curves

Unit 5: Parametric surface modeling

Parent-Child Relationships, Nesting; Feature Arrays; Nesting; Design Variations;

Unit 6: Fundamentals of Class A surface modeling

G0, G1, G2, G3 Continuity, Optical Analysis of Multi-span surfaces, Curvature Analysis

References

Christoph M. Hoffmann, Geometric and Solid Modelling: An Introduction
William Howard and Joseph Musto. Introduction to Solid Modeling Using solid works. McGraw Hills
Alejandro Reyes, Beginners Guide to SolidWorks, SDC Publications

ADS271 –DESIGN DOCUMENTATION - I

A craft/cottage Industry

Understanding of the current and past scenario of craft and cottage industries of Uttar Pradesh or any other state of India.

Field study of selected craft and cottage industry and pinpoint lacuna in their process/artifacts etc.

Course Outcome

Comprehensive understanding of the history and present state of one cottage industry/craft of Uttar Pradesh or India

Aptitude to conduct thorough field research and collect qualitative and quantitative information which may be useful for other industries and people

ADS281 –DESIGN PROJECT–I, SIMPLE PRODUCT DESIGN

Lectures on simplicity; Complex nature of simple products; Single Function Products; Material Property as function; Form as function.

Course Outcome

Ability to create innovative single component single feature solutions

ADS221–DESIGN MANAGEMENT–I, CONSUMER PSYCHOLOGY

Course Outcome

Understating of why people buy things

Ability to offer relevant products which people need

Awareness about Design Rights and their enforcement by law

Unit 1: Consumer Motivations

Identification of user needs and Driving Factors;

Emotional Design, Sensibility, Social Ethics and Concerns;

Unit 2: Market

Consumer Vs Buyer, Consumer Groups, Buyer Groups, Periodic Trends,

Market Gaps, Market Oriented Innovation;

Unit 3: Business Evolution

Product Planning for the future, Disruptive Innovation;

Unit 4: Product Lifecycle Management

Procurement, Process monitoring, Quality Assurance, Guarantee Statement, Warrantee Statement, Buybacks, refurbishing;

Component interchangeability, Process Homogenization, Material Homogenization, Feature Standardization;

Unit 5: Intellectual Property

Product differentiation and identity;

Patent laws, History, Budapest Treaty, Current State of Enforcement; Structure of a Patent Application;

Design Rights, Trademarks, and intellectual property, India Design Act;

Global and local frameworks of securing Intellectual Property Rights; indigenous intellectual property;

Indian laws and their enforcement mechanisms. International laws and their enforcement mechanism;

References

Kathryn Best, The Fundamentals of Design Management, AVA Publishing, 2010

Brigitte Borja De Mozota, Design Management: Using Design to Build Brand Value and Corporate Innovation, Allworth Press, 2004

Kenneth B Khan, Product Planning Essentials, M E Sharpe Inc, 2011

John Stark, Product Lifecycle Management: 21st Century Paradigm for Product Realisation, Springer, 2011

Craig M. Vogel and Jonathan Cagan, Creating Breakthrough Products: Innovation from Product Planning Program Approval, FT Press, 2001

David L. Rainey, Product Innovation: Leading Change through Integrated Product Development, Cambridge University Press, 2011

ADS222–DESIGN THINKING

Course Outcome

Ability to comprehend large scale problems.

Ability to come up with radical solutions in relatively short time.

Unit 1: Intention

Understanding Empathy; User stories; interpretive research;

Unit 2: Problem Definition

Visual representation of problem statements; contextualization and validation of a problem set; Heat maps; polar maps; Data visualization methods;

Unit 3: Scenario Building

Mapping solutions; partial solutions; incremental solutions

Unit 4: Ideation

Brainstorming; Differential Discussion; group methods to generate ideas; solitary methods to generate ideas; Lateral Thinking

Unit 5: Concept Detailing

User Journey maps; User stories; activity mapping; feature matrix

References

John Thackara, In the Bubble: Designing in a Complex World, The MIT Press, 2005
Bruce Hanington, Bella Martin, Universal Methods of Design: 100 Ways to Research Complex Problems, Develop Innovative Ideas, and Design Effective Solutions, Rockport Publishers, 2012
Donald A. Norman, Living with Complexity, MIT Press, 2010
Jeffrey Whitten and Lonnie Bentley, Systems Analysis and Design Methods, McGraw-Hill/Irwin, 2005
Gerald M. Weinberg and Daniela Weinberg, General Principles of Systems Design, Dorset House, 1988

ADS223–SOLID MODELING IN COMPUTER AIDED DESIGN

Unit 1: Solid Modeling basics

Linear Extrusion; Boolean operations;

Unit 2: Parametric Solid modeling

Surface handling; Curvature handling; thickness handling

Unit 3: Assembly

Skeleton modeling; DOF restriction; reference matching; tolerances;

Unit 4: Product integration in CAD

Bottom-Up Modeling; Top-Down Modeling; Fasteners;

Unit 5: Variation

Design explorations

Unit 6: Drafting

Drafting and generating 2D drawings from 3D models

Generation of Bill of materials

Course Outcome

Ability to finalize component level design

Ability to improvise a design before making physical model

Ability to simulate and analyse a solution

Ability to furnish production details

References

Christoph M. Hoffmann, Geometric and Solid Modelling: An Introduction
William Howard and Joseph Musto. Introduction to Solid Modeling Using solid works. McGraw Hills
Alejandro Reyes, Beginners Guide to SolidWorks, SDC Publications

ADS261 –NATURE AND FORM

Form and metaphors

Inspirations from nature, Utility, Evolution

Biomimicry, Biosimilar shapes and volumes, Exoskeletons, Structures

Bionics, Biosimilar components of mechanisms

Applications in product Design

3D modeling

Course Outcome

Ability to take inspiration from nature to solve product design trials

References

Maggie Macnab, Design by Nature: Using Universal Forms and Principles in Design, New Riders, 2011
Rudolf Finster walder, Form Follows Nature: A History of Nature as Model for Design in Engineering, Architecture and Art, Springer Vienna Architecture, 2011
Alan Powers, Nature in Design: The Shapes, Colors and Forms that Have Inspired Visual Invention, Conran, 2002
Ellen Lupton, Jennifer Tobias, Alicia Imperiale, Grace Jeffers and Randi Mates, Skin: Surface, Substance, and Design, Princeton Architectural Press, 2002

ADS291 –DESIGN PROJECT–II, DISPLAY AND CONTROL DESIGN

Course Outcome

Practical Application of Cognitive and Physical Ergonomics

Understanding Mental models

Understanding how people react with digital devices

Orientation: Cognitive Ergonomics

Unit 1: Morphology

Morphology of Interactive media; Data; Consumer; Input Channels; Response mechanisms; gratification models; Activity mapping; Stimulus-Response; Action-Reward;

Unit 2: Narrative

Storytelling and narratives in Interactive media

Unit 3: User Experience

Experience design and interactive media

Unit 4: Information Architecture

Information Structuring, time and space for Interactive Medias

Unit 5: User Interface Design

Design of multi-modal interfaces for text, graphics, animation, video, audio, games, etc.

Project Work

Designing interactive media for public use - installations, Museums and community facilities

References

Lev Manovich, The Language of New Media, Cambridge, MIT Press, 2001
Joe Lambert, Digital Storytelling: Capturing Lives, Creating Community, Life on the Water Inc, 2008
Noah Wardrip- Fruin (Editor) and Nick Montfort, The New Media Reader, MIT Press, 2003
Bryan Alexander, The New Digital Storytelling: Creating Narratives with New Media, Praeger, 2011
Robert Klanten, Interactive Installations and Experiences, Die Gestalten Verlag, 2011

ADS311 –CREATIVE NARRATION

Course Outcome

Ability to comprehend and draft user stories.

Ability to document user experience in a retainable form.

Unit 1: Story

Story, narrative and meaning making, metaphors

Unit 2: Objective

Premise and problem statement, Context

Unit 3: Protagonist

Characters and personas, examples of Don Quixote, Karna

Unit 4: Chain of Events

Plot and Scenarios

Unit 5: Dynamics

Relationship between problems, need and conflict, rationalization of need, rationalization of conflict

Unit 6: Closure

Conflict, Action and Resolution

References

Mike Korolenko and Bruce Wolcott, Storytelling and Design: Media Literacy for the Digital Age, Pearson Learning Solutions, 2005
Marie-Laure Ryan (editor), Narrative across Media: The Languages of Storytelling, University of Nebraska Press, 2004
Kristin M. Langellier and Eric E. Peterson, Storytelling In Daily Life: Performing Narrative, Temple University Press, 2004

ADS312–HUMANITIES & SOCIAL STUDIES

Unit 1: Artistic Creation

Complete understanding of Perception, Communication, Imagination, Expression, and Creativity for artistic creation.

Unit 2: Art & Design Movements

To understand the thoughts and techniques involved in important art movements - Impressionism, Cubism, Constructivism, Optical Art, Kinetic Art etc. Students are expected to express their ideas through Posters, Murals, Building Art, Collage, Graffiti, 3D-Installations.

Unit 3: Ethnography and Anthropology

Observations/Analysis; Community---Based Ethnographic Research; Activity Theory; Empathy in Design; Value Sensitive Design; Historical development of fieldwork; relations between field methods and dominant theoretical orientations; varieties of fieldwork at present; the implications; Ethnographic research design as a continuous process; the formulation of research problems.

Unit 4: Introductory Sociology & Psychology

Sociology as a Science of Human Society: Introduction: - Basic concepts (Roles, Norms, Values, Groups and Institutions), Social Structure, Culture, Perspectives (Functionalist, Conflict & Interactionist), Psychological Perspectives and Approaches;

References

Geraldine Gay and Helene Hembrooke, 2004, Activity-Centered Design, An Ecological Approach to Designing Smart Tools and Usable Systems.
Amy. E. Aniston, Graphic Design Basics (IInd Edition)
Lydia Darbyshire, Practical Graphic Design Technique
Batya Friedman and Alan Borning, Value Sensitive Design and Information Systems
Julian Murchison, Ethnography Essentials: Designing, Conducting, and Presenting Your Research

Course outcome

Develop visual awareness of the present day environment.

Recognize and relate design forms to historical precedents and possible future developments.

Have a greatly increased general knowledge, including source material which he/she can draw on for future student and development in the field of design

Producing reliable information about human, social life and culture.

ADS313–DESIGN MANAGEMENT–II, BUSINESS COMMUNICATION

Course Outcome

Ability to communicate novel ideas to general stake holders

Unit 1: Explaining Ideas
Briefs, Detailed Briefs and Concept Notes

Unit 2: Scale of Work
Approximation, Cost Estimation

Unit 3: Discussion
Knowledge Management, Feedbacks, Information loops, Updates

Unit 4: Record Keeping
Activity Documentation and Reporting

Unit 5: Scope of Work
Induction, Work distribution, Task Assignments

Time management, Scheduling

Unit 6: Presentation
Informal and Formal Communication; Top-Down, Bottom-Up, Diagonal and Lateral Communication; Pitch Presentations.

References

Best, K. (2006). Design management: managing design strategy, process and implementation. AVA publishing
Cooper, R., Junginger, S., & Lockwood, T. (Eds.). (2013). The handbook of design management. A&C Black
Martin, R. L. (2009). The design of business: Why design thinking is the next competitive advantage. Harvard Business Press.

ADS314–DESIGN RESEARCH METHODOLOGY–I, QUANTITATIVE RESEARCH

Course Outcome

Ability conduct objectivist research

Ability to understand large scale data

Ability to understand collective behaviour

Unit 1: Elements of Research

Data; Variables and constants; Concepts and constructs; Theories and observations;

Unit 2: Logic

Inductive; Deductive;

Unit 3: Sampling

Basic concept; Representativeness; Probability and non-probability sample- concepts; Types of probability and non-probability samples; Practical guidelines for sampling

Unit 4: Quantitative Methods and Statistics

Surveys: importance, types, uses, guidelines for framing questionnaire; Content Analysis- basic concepts, Characteristics and uses; Distribution;

Unit 5: Statistics in media research

Measures of Central tendency; Descriptive statistics, Correlations, Inferential statistics- chi square, t – test

Unit 6: Field Work

Taking up a topic and collecting data from locations; Equivalent to 8 lecture hours;

References

Brenda Laurel, Design Research: Methods and Perspectives, The MIT Press, US, 2003

R. D. Wimmer & J. R. Dominick, Mass media research: An introduction. Belmont, California, Wadsworth Pub. Co., 2000

A. Hansen, Mass communication research methods. New Delhi: Log Angeles, 2009

E. R. Babbie, The practice of social research. Belmont, California, Wadsworth Pub. Co., 1992

C. R. Kothari, Research methodology: Methods & techniques. New Delhi: New Age International (P) Ltd., 2004

ADS315 –ARCHITECTURAL STUDIES–II, CONNECTIVITY AND MOBILITY

Course Outcome

Understanding of how humans address their need to interact with others as social animals

Unit 1: Connectivity

Communication channels; interpersonal communication; mass communication; Public Information systems; Information Services;

Information Consumption patterns; Internet of Things;

Unit 2: Movement

Movement of Goods; Movement of People; Modes of Transportation; Individual Movement; Mass movement; Transportation as service; Transit Systems; Commuter centric Design;

Public Vs Individual information systems; Mobile navigation; mental maps; Traffic Management;

Unit 3: Engagement

Public gatherings; Public events; Event management and planning; Crowd Management;

Unit 4: Infrastructure

Resources sharing and management; Delivery systems; Consumption patterns; Occupancy patterns;

Remote Area Connectivity; Rural Transportation; Rural Road networks;

Urban Connectivity; High Density Transportation; Transit Oriented Development; Public Safety;

Accessible Design; Universal Design in Public Transportation; Universal Design in Communication technologies;

Unit 5: Sustainable Transportation

Structure and design flaws of Public Transportation; Alternative Energy resources for transportation; Anatomy of Electric Vehicles;

References

Caspers, Mar, Designing Motion, Automotive Designers 1890 to 1990, Birkhäuser, 2017
Brandes, Uta / Stich, Sonja / Wender, Mir, Design by Use, The Everyday Metamorphosis of Things, Birkhäuser, 2009
Alice Foxley, Distance and Engagement: Walking, Thinking and Making Landscape, Lars Müller, 2001
Kevin Thwaites. Sergio Porta, Urban Sustainability Through Environmental Design: Approaches to Time-People-Place Responsive Urban Spaces, Taylor & Francis, 2007
Herwig, Oli, Universal Design, Solutions for Barrier-free Living, Birkhäuser, 2008

ADS351 – DESIGN WORKSHOP

Course Outcome

Hands on experience …

Emphasis on the skill of workshop methods. Hands on techniques to manipulate the basic materials according their understanding. Exposure to the students on some of the software on Product Design and Visual Design. The focus of the course is teaching the students about design intent and how software can be utilized for the maximum benefit of the designer. The student project involves making products out of concepts in virtual environment.

References

Williams, R., & Sheldon, C. Robin Williams Handmade Design Workshop: Create Handmade Elements for Digital Design. Peachpit Press., 2009

Cuffaro, D., & Zaksenberg, I. The Industrial Design Reference & Specification Book: Everything Industrial Designers Need to Know Every Day. Rockport Publishers.2010

Sherwin, D.Creative workshop: 80 challenges to sharpen your design skills. How Books.2013

Fullerton, T., Swain, C., & Hoffman, S. Game design workshop: Designing, prototyping, & playtesting games. CRC Press. 2004

ADS381 –DESIGN PROJECT – III, SOFTWARE USER INTERFACE DESIGN

Course Outcome

Ability to design compelling interfaces of software products

Orientation

Unit 1: Theory

Use of HCI methods (Contextual Enquiry, Focus Groups, Interviews, etc.) for understanding the user and his requirement.

Unit 2: Research

Understanding the factors that define user experience.

Unit 3: Design

Design of multi-modal interfaces, expressive interfaces, audio interfaces, tangible interfaces and gestural interfaces.

Design of interactive systems, products for future use, Collaborative products to be used in groups, devices for rural applications and devices for use in public places

References

Donald A. Norman, Invisible Computer: Why Good Products Can Fail, the Personal Computer Is so Complex and Information Appliances Are the Solution, MIT Press, 1998
Brenda Laurel, Computer as Theater, Addison-Wesley Pub Co, 1993
Jef Raskin, The Humane Interface: New Directions for Designing Interactive Systems, Pearson Education,2000

ADS321 –PRODUCT BRANDING AND IDENTITY

Course Outcome

Understanding of why organizations invest in maintaining their identity

Knowledge of Branding and Identity design process

Unit 1: Principles of Visual Design Refresher

Understanding visual culture; Visual Theories; Visual Design; Symbolism, Time, Sound; Point of View

Unit 2: Visual Art

Visual art History; Painting; Architecture and Sculpture; Artistic Styles;

Unit 3: Aesthetic Experience

Modes of Aesthetic Experience; Basics of Aesthetic values; Aesthetics of Thinking and Creativity; Taste and Aesthetes; Aesthetics of Symbols and Language;

Unit 4: Visual Experience

Photography and Moving Images; Historical, Technical and Cultural Perspective; Ethical and Critical Perspective; Motion Pictures; Television and Video; Reality Shows;

Unit 5: Branding and Identity

History of branding; structure of a Brand; Brand language; Logos; Copywriting; Typeface; Brand Placement; Brand Guidelines;

Structure of Identity; Visual Abstraction; Metaphors; Communication; Representativeness; Evolution;

References

Paul M., Visual Communication: Images with Messages, 2006
Ralf E. Wileman, Visual Communication
David Sless, Learning & Visual Communication
Friedrich O. Huck and Carl L. Fales, The Digital Evolution: Visual Communication in the electronic age
Zia-Ur-Rehman, Visual Communication an Information Theory Approach
Margaret Mark, Carol Pearson, the Hero and the Outlaw: Building Extraordinary Brands through the Power of Archetypes, McGraw Hill, 2001

ADS322–DESIGNRESEARCH METHODOLOGY–II, APPLIED STATISTICS AND INFOGRAPHICS

Course Outcome

Ability to understand and communicate complex information in a consumable way

Ability to find relationships between statistical data accumulated from large scale surveys

Unit 1: Regression models

Linear regression, Simple regression, Ordinary least squares, Polynomial regression, General linear model;
Generalized linear model, discrete choice, Logistic regression, Multinomial logit, mixed logit, Probit, Multinomial probit, Ordered logit, Ordered probit, Poisson;
Multilevel model, fixed effects, Random effects, Mixed model;

Unit 2: Nonlinear Regression

Nonlinear regression, Nonparametric, Semiparametric, Robust, Quantile Isotonic, Principal components, Least angle, Local, Segmented, Errors-in-variables;

Unit 3: Statistical Extrapolation

Probability Distribution; Linear extrapolation; Polynomial extrapolation; Conic Extrapolation; French Curves;
Least squares, Ordinary least squares, Linear, Partial, Total, Generalized, Weighted, Non-linear, Non-negative, Iteratively reweighted, Ridge regression, Least absolute deviations, Bayesian, and Bayesian multivariate Linear Regression.
Statistical Forecasting; Average, Naïve, and seasonal Naïve approaches, Drift method;
Qualitative forecasting methods, informed opinion and judgment, Delphi method, market research, and historical life-cycle analogy;

Quantitative Forecasting methods, Time Series methods, last period demand, simple and weighted N-Period moving averages, simple exponential smoothing, poisson process model based forecasting, and multiplicative seasonal indexes; Extreme Value Theory

Interpretive approach: Cool hunting; Dead Reckoning;

Unit 4: Data Visualization

Data Point, Bar, Stack, Pie, Donut, and Dot charts; Histograms;

Unit 5: Graphic Design and Data Visualization

Infographics Design project

References

Brenda Laurel, Design Research: Methods and Perspectives, The MIT Press, US, 2003
R. D. Wimmer & J. R. Dominick, Mass media research: An introduction. Belmont, California, Wadsworth Pub. Co., 2000
A. Hansen, Mass communication research methods. New Delhi: Log Angeles, 2009
E. R. Babbie, The practice of social research. Belmont, California, Wadsworth Pub. Co., 1992
C. R. Kothari, Research methodology: Methods & techniques. New Delhi: New Age International (P) Ltd., 2004

ADS323–MATERIAL AND PROCESSES IN DESIGN–II, UNCONVENTIONAL MANUFACTURING

Course Outcome

Ability to design products which are seemingly impossible to manufacture using conventional processes

Unit 1: Material Deposition Processes

Laser Deposition, Micro-Plasma Powder Deposition, Chemical vapor Deposition, Micro Welding, Powder Casting

Metal 3D Printing, Powder Deposition 3D printing;

Unit 2: Subtractive Processes

Electrochemical machining, Electro-Discharge machining, Ultrasonic Machining, Laser Beam Machining, Water jet machining, Abrasive Jet Machining, Plasma Arc machining

Unit 3: Cutting and Removal

Water Jet Cutting, Plasma Cutting, Laser Cutting, Electro-Discharge Wire Cutting; Abrasive Jet Cutting

Unit 4: Additive Extrusion Processes

Extruded Filament 3D printing, Clay 3D printing, Stereo lithography

Unit 5: Special Purpose Manufacturing processes

Rot molding, Layer Compression, Sheet contouring, Friction Welding

Unit 6: Surface Treatment Processes

Laser Etching, Acid/Base Etching, Electro Chemical Etching, Sand Blast Etching, Ultraviolet Etching, Photochemical Machining

Electro Chemical Polishing

References

Vijay K. Jain, Advanced Machining Processes. Allied Publishers, New Delhi, 2007
P. C. Pandey and H.S. Shan, Modern Machining Processes, Tata McGraw-Hill, New Delhi, 2007
G.F. Benedict, Nontraditional Manufacturing Processes, Marcel Dekker Inc., New York, 1987
McGeough, Advanced Methods of Machining, Chapman and Hall, London, 1998
Paul De Garmo, J.T. Black, and Ronald A. Kohser, Material and Processes in Manufacturing, Prentice Hall India, 2001.

ADS361 –CONTROL SYSTEMS IN DESIGN

Course Outcome

Theoretical awareness of how control systems of products work

Unit 1: Sensing

Feedback systems, mathematical modelling of physical systems;

Unit 2: Signals

Laplace transforms, block diagrams, signal flow graphs, state-space models;

Unit 3: Data

Time domain analysis: performance specifications, steady state error, transient response of first and second order systems;

Unit 4: Processing

Stability analysis: Routh-Hurwitz stability criterion, relative stability; proportional integral, PI, PD, and PID controllers;

Lead, lag, and lag-lead compensators;

Unit 5: Analysis

Root-locus method: analysis, design;

Frequency response method: Bode diagrams, Nyquist stability criterion, performance specifications, design; State space methods: analysis, design; Physical realizations of controllers: hydraulic, pneumatic, and electronic controllers.

References

K Ogata, Modern Control Engineering, 4th Ed., Pearson Education Asia, 2002
B C Kuo and F. Golnaraghi, Automatic Control Systems, 8th Ed., John Wiley (students ed.), 2002
M Gopal, Control Systems: Principles and Design, 2nd Ed., TMH, 2002
M Gopal, Modern Control System Theory, 2nd Ed., New Age International, 1993
R. C. Dorf and R. H. Bishop, Modern Control Systems, 8th Ed., Addison Wesley, 1998
P. Belanger, Control Engineering: A modern approach, Saunders College Publishing, 1995

ADS391 –DESIGN PROJECT–IV, TECHNICALLY COMPLEX PRODUCT DESIGN

Course Outcome

Ability to break down complex product ideas into smallest components

Ability to build functional prototypes

Orientation

Product Integration; Forward Integration; Backward integration;

Top – Down Design Approach; Bottom-Up Design Approach;

Reverse Engineering; Proof Of concept;

DS411–PROFESSIONAL PRACTICE IN DESIGN

Course Outcome

Understanding of how modern businesses start and run.

Ability to supervise and manage live design projects

Basic skillset of a design business owner

Unit 1: Entrepreneurship

Social Entrepreneurship, Business Entrepreneurship, Trading Entrepreneurship, Corporate Entrepreneurship, and Agricultural Entrepreneurship;

Unit 2: Business Foundation

Timmons Model of Entrepreneurship, Investment Models, Startup Business Models, Business Plans, Pitch presentations, Small Business models;

Unit 3: Legal aspects of business

Contracts and Agreements, Conflict Resolution, Arbitration;

Unit 4: Running a Design Business

Set up of an independent design business, Hiring processes, Project Scheduling and work delegation, Cost Estimation;

Unit 5: Intellectual Property

Registration process of Intellectual property Rights, Design Registration, and Brand/Trademark Registration;

Unit 6: Accounting

Billing, salaries and taxation;

References

Ted Crawford, AIGA Professional Practices in Graphic Design, Allworth Press, 2008
Shan Preddy, How to Run a Successful Design Business: The New Professional Practice, Gower Publishing, Ltd., 2011

ADS412 – FORM IN FOUR DIMENSIONS

Course Outcome

Understanding of evolution of shapes and forms with time

Unit 1: Projection and Transition

Movement of Light around a Form, Movement in time and space. Movement of Space around a Form

Kinematic Structures, Mechanisms, spontaneous, automatic and reactive dynamic forms

Unit 2: Transformation

Fluid Forms, Semi Fluid Forms, Semi Solid Forms

Unit 4: Evolution

Feature based evolution; Content based evolution; Environment Based Evolution

Unit 5: Union and Separation

Merger, Conformity, Subtraction, Division

Unit 6: Decay

Concept of Decay, Examples from nature, examples of manmade objects

References

Ellen Lupton, Jennifer Tobias, Alicia Imperiale, Grace Jeffers and Randi Mates, Skin: Surface, Substance, and Design, Princeton Architectural Press, 2002

Li: Dynamic Form in Nature, Wooden Books, 2007

Lung-Wen Tsai, Mechanism Design: Enumeration of Kinematic Structures According to Function, CRC Press, 2001

Sibel Deren Guler, Madeline Gannon, KateSicchio, Crafting Wearables: Blending Technology with Fashion, 2016

ADS413–MATERIALS AND PROCESSES–III, PRODUCTION PLANNING

Course Outcome

Understanding manufacturing and mass production

Understanding a designer's role to facilitate manufacturing

Ability to innovate for cost and carbon footprint reduction of mass produced goods

Unit 1: Fundamentals of Mass Production

Introduction, Production Planning and Control, Value analysis and value engineering

Unit 2: Structure of Mass Production

Plant location and layout, Equipment selection, Maintenance planning

Job, batch, and flow production methods

Group technology, Work study, Time and motion study, Incentive schemes, Work/job evaluation, Inventory control,

Unit 3: Manufacturing planning

MRP, MRP-II, JIT, CIM, Quality control, Statistical process control, Acceptance sampling, Total quality management, Taguchi's Quality engineering.

Forecasting, Scheduling and loading, Line balancing, Break-even analysis.

Unit 4: Operations

Introduction to operations research, linear programming, Graphical method, Simplex method, Dual problem, dual simplex method, Concept of unit worth of resource, sensitivity analysis, Transportation problems, Assignment problems

Network models: CPM and PERT, Queuing theory

Unit 5: Alternatives of Mass production

Studio hours to be committed to explore 3D printing and additive manufacturing

References

S. L. Narasimhan, D. W. McLeavey, and P. J. Billington, Production, Planning and Inventory Control, Prentice Hall, 1997
J. L. Riggs, Production Systems: Planning, Analysis and Control, 3rd Ed., Wiley, 1981
Muhlemann, J. Oakland and K. Lockyer, Productions and Operations Management, Macmillan, 1992
H. A. Taha, Operations Research - An Introduction, Prentice Hall of India, 1997
J. K. Sharma, Operations Research, Macmillan, 1997.

ADS471 –INDUSTRIAL TRAINING

At a sponsor business

Students in this semester would avail opportunity to work with design firm under the guidance of practice professional for 6-8 weeks.

Course Outcome

Work Experience under the guidance of in practice professionals

ADS472 –DESIGN DOCUMENTATION - II

Location and Industry/Craft

Understanding of the history and present state of one cottage and craft industry of UP. To understand the conventional processes, materials and prepare a product catalog, Design Insights and proposals for survival and advancement of the subject industry/craft

Course Outcome

Comprehensive understanding of the history and present state of one cottage industry/craft of Uttar Pradesh or India

Design inputs for the benefit of the targeted craft/cottage industry

Aptitude to conduct thorough field research and collect qualitative and quantitative information which may be useful for other industries and people

ADS481 – DESIGN PROJECT – V, SYSTEM THINKING IN DESIGN

Course Outcome

Ability of comprehend large scale problems

Skillset to work in the collaborative dynamic of a design team

Development of systemic approach to solve ill-defined problem

Social/environmental awareness

Unit 1: System Thinking

Lectures on Design Thinking and System Thinking from Design perspective. The Fifth Discipline approach. Scenario Maps and Metaphors

Unit 2: Problem Solving

Design of system level solutions so that design can be thought of modularly as suited for different combinations and applications.

Unit 3: Complex Systems

Understanding, strategizing, conceptualizing and designing for complex systems.

Unit 4: System Design

Designing complex artefacts. Design solutions that are suitable for transportation, education, publishing, retailing, etc.

References

John Thackara, In the Bubble: Designing in a Complex World, The MIT Press, 2005
Bruce Hanington, Bella Martin, Universal Methods of Design: 100 Ways to Research Complex Problems, Develop Innovative Ideas, and Design Effective Solutions, Rockport Publishers, 2012
Donald A. Norman, Living with Complexity, MIT Press, 2010
Jeffrey Whitten and Lonnie Bentley, Systems Analysis and Design Methods, McGraw-Hill/Irwin, 2005
Gerald M. Weinberg and Daniela Weinberg, General Principles of Systems Design, Dorset House,1988

ADS499 –DESIGN DEGREE PROJECT

Sponsored/Self Sponsored full time Project

Students in this semester would take sponsored /self-sponsored fulltime project (technically complex project). The students can do design degree project in house but it would be advisable for them to go to industry/ design firm and do the design degree project

ELECTIVES

ADS331- Elective I

ADS331a Sustainable Design

ADS331b Furniture & Interior Design

ADS331c Frugal Innovation

ADS332- Elective II

ADS332a Design of Medical Equipment

ADS332b Mobility and Vehicle Design

ADS332c Aerospace Design

ADS341- Elective III

ADS341a Videography and Film Making

ADS341b Graphics and Animation

ADS341c Typography

ADS342- Elective IV

ADS342a Interaction Design

ADS342b Universal Design

ADS342c Toy and Games Design

ADS431- Elective V

ADS431a Photography and Image Processing

ADS431b FEM and Optimization Tools

ADS431c Mechatronics

Course Outcome

To make students specialized in particular domains

EVALUATION SCHEME

SEMESTER I

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	ADS 101	Introduction to Design	1	0	6	30	70	100	0	50	150	**4**
2	ADS 102	Representation Techniques	0	0	4	15	35	50	0	50	100	**2**
3	ADS 103	Elements of Design	1	0	8	30	70	150	0	50	200	**5**
4	ADS 104	Applied Sciences for Design	2	0	6	15	35	150	0	50	200	**5**
5	ADS 105	Tinkering Studio	1	0	6	30	70	100	0	50	150	**4**
6	AME 111	Engineering Drawing	1	0	2	15	35	50	50	0	100	**2**
7	AEE 101	Electrical Sciences	3	0	2	15	35	50	50	0	100	**2**
		Total	**9**	**0**	**30**						**1000**	**24**

SEMESTER II

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	ADS 111	Form, Order and Structure	1	0	8	30	70	100	0	100	200	**5**
2	ADS 112	Principles of Visual Design	1	0	8	30	70	100	0	100	200	**5**
3	ADS 113	Visualization and Illustration	0	0	10	30	70	100	0	100	200	**5**
4	ACS 101	Introduction to Computing	3	0	0	15	35	50	50	0	100	**3**
5	ACS 110	Computing Laboratory	0	0	4	15	35	50	0	50	100	**2**
6	AEE 102	Basic Electronics Laboratory	0	0	4	15	35	50	0	50	100	**2**
7	AME 101	Engineering mechanics	1	0	2	15	35	50	50	0	100	**2**
		Total	**8**	**0**	**28**						**1000**	**24**

SEMESTER III

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	ADS 211	Architectural Studies in Design - I	2	0	4	30	70	100	0	100	200	**4**
2	ADS 212	Communication Studies in Design- II	3	0	0	30	70	100	0	50	150	**3**
3	ADS 213	Model Making and Hand Tools Workshop	1	0	6	30	70	100	0	50	150	**4**
4	ADS 214	Surface Modelling in CAD	1	0	8	30	70	100	0	50	150	**3**
6	RVE 301	Universal Human Values and Professional Ethics	3	0	0	15	35	50	50	0	100	**3**
7	ADS 271	Design Documentation – I	0	0	0	0	0	0	0	50	50	**2**
8	ADS 281	Design Project – I	0	0	10	0	100	100	0	100	200	**5**
9	---	Cyber Security	2	0	0							**0**
		Total	**12**	**0**	**28**						**1000**	**24**

SEMESTER IV

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	ADS 221	Design Management - I	2	0	2	30	70	100	0	50	150	**3**
2	ADS 222	Design Thinking	1	0	6	30	70	100	0	100	200	**4**
3	ADS 223	Solid Modelling in CAD	1	0	6	30	70	100	0	50	150	**4**
4	---	English Language and Technical Writing	2	0	2	15	35	50	50	0	100	**3**
5	ADS 261	Nature and Form	0	0	10	30	70	100	0	100	200	**5**
6	ADS 291	Design Project – II	0	0	10	0	100	100	0	100	200	**5**
7	---	Environmental Science	2	0	0							**0**
		Total	**8**	**0**	**36**						**1000**	**24**

SEMESTER V

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	ADS 311	Creative Narration	2	0	0	15	35	50	0	50	100	2
2	ADS 312	Humanities and Social Studies	2	0	0	15	35	50	50	0	100	2
3	ADS 313	Design Management – II	2	0	0	15	35	50	0	50	100	2
4	ADS 314	Design Research methodology - I	2	0	0	15	35	50	50	0	100	2
5	ADS 315	Architectural Studies in Design – II	1	0	4	15	35	50	0	50	100	3
6	ADS 331	Elective – I	1	0	2	15	35	50	0	50	100	2
7	ADS 332	Elective – II	1	0	2	15	35	50	0	50	100	2
8	ADS 351	Design Workshop	0	0	8	15	35	50	0	50	100	4
9	ADS 381	Design Project - III	0	0	10	0	100	100	0	100	200	5
10	---	Constitution of India	2	0	0							0
		Total	**13**	**0**	**26**						**1000**	**24**

SEMESTER VI

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	ADS 321	Product Branding and Identity	2	0	4	30	70	100	0	50	150	**4**
2	ADS 322	Design Research methodology - II	3	0	2	30	70	100	0	100	200	**4**
3	ADS 323	Materials and Processes in Design - I	1	0	8	30	70	100	0	50	150	**5**
6	ADS 341	Elective - III	1	0	2	15	35	50	0	50	100	**2**
7	ADS 342	Elective - IV	1	0	2	15	35	50	0	50	100	**2**
8	ADS 361	Control system in Design	0	0	4	15	35	50	0	50	100	**2**
9	ADS 391	Design Project – IV	0	0	10	0	100	100	0	100	200	**5**
10	---	Essence of Indian Traditional Knowledge	2	0	0							**0**
		Total	**10**	**0**	**32**						**1000**	**24**

SEMESTER VII

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	ADS 411	Professional Practice in Design	2	0	0	30	70	100	0	50	100	**2**
2	ADS 412	Form in Four Dimensions	2	0	10	30	70	100	0	100	200	**5**
3	ADS 413	Materials and Processes in Design - II	2	0	2	30	70	100	0	50	150	**3**
4	ADS 431	Elective - V	1	0	2	15	35	50	0	50	100	**2**
6	ADS 471	Industrial Training	0	0	0	0	0	0	0	150	150	**3**
7	ADS 472	Design Documentation-II	0	0	0	0	0	0	0	100	100	**4**
8	ADS 481	Design Project - V	0	0	20	0	100	100	0	100	200	**5**
		Total	**7**	**0**	**34**						**1000**	**24**

SEMESTER VIII

Sl. No.	Course Code	Subject	Periods			Internals			End Semester		Total	Credit
			L	T	P	CT	TA	Total	TE	PE		
1	ADS 499	Design Degree Project	0	0	48	0	500	500	0	500	1000	**24**
		Total	**0**	**0**	**48**						**1000**	**24**